

Spectra-FakeNet: A Dual-Stream Spatial-Frequency Attention Network for Robust Deepfake Detection

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Abstract. The rapid progress of deep generative models has made face manipulation, commonly known as deepfakes, cheap to produce and increasingly difficult to tell apart from genuine footage by eye. Most existing detectors are convolutional networks trained purely on RGB pixels, and while they report high accuracy on the benchmark they were trained on, that accuracy tends to fall away sharply once the network is shown a manipulation method or a compression level it has not seen before. This paper works from a recent survey of the field to identify why: spatial-only detectors overfit to the specific pixel artifacts left by one generator and have no mechanism for picking up the statistical residue that generative adversarial networks leave in the frequency domain. We propose Spectra-FakeNet, a dual-stream network that processes a face crop as two parallel signals, an RGB stream that behaves like a conventional forensic CNN and a Discrete Cosine Transform stream that looks at the log-magnitude spectrum of the same crop for the grid-like up-sampling artifacts that are invisible to the spatial stream. The two streams are fused through a concatenation and dense layer before a sigmoid classifier makes the real/fake decision. On our evaluation set, the fused model reached perfect separation between real and fake classes and, under Gaussian blur, held 92% accuracy against 78% for a spatial-only baseline, and it correctly flagged the majority of fakes that passed a purely visual check. We report these results together with their limits, since the evaluation set is small and drawn from a single source, and lay out what a larger cross-dataset study would need to look like.

Keywords. deepfake detection, frequency domain analysis, discrete cosine transform, dual-stream convolutional networks, GAN artifact detection, media forensics.

1. Introduction

Face-swapping and face-reenactment tools have gone from research demos to something almost anyone can run on a laptop or a phone. The consequence is a steady rise in manipulated video and imagery circulating on social platforms, and with it a corresponding rise in the cost of not being able to tell real footage from synthetic footage. Detection research has largely converged on convolutional neural networks trained end to end on RGB frames, and these networks are genuinely good at this task when the training and test data come from the same generator and the same compression pipeline.

The trouble is that this is rarely the situation a detector meets once it is deployed. A model trained on one face-swap tool sees its accuracy fall when it is shown output from a different one, and it falls further once ordinary social-media compression is applied, because the fine spatial artifacts the model learned to key on are exactly the kind of detail that compression removes first. This is a generalisation problem, not an accuracy problem, and it is the reason a detector can look excellent on a leaderboard and still be unreliable in practice.

This project starts from a 2025 Springer survey on multimedia-enabled deepfake detection, which reviews detection work from 2020 to 2025 and states this generalisation gap plainly: models such as XceptionNet and MesoNet reach above 98% accuracy on clean benchmark data such as FaceForensics++, but overfit to the specific artifact signature of their training generator and degrade on unseen manipulation types or compressed video [1]. The survey also points to a gap in the literature it reviews, namely that very few detectors make use of the frequency domain, even though up-sampling layers inside GAN generators are known to leave a periodic, grid-like signature in an image’s spectrum that is largely invisible in the spatial domain [6, 7].

That gap is the starting point for this paper. Rather than treating frequency analysis as a separate, hand-crafted foren-

sic check, we build it directly into the network as a second learned stream that runs in parallel with a conventional spatial branch, so the model can weigh visual plausibility against spectral plausibility for every image it sees.

This paper makes the following contributions:

1. We identify, following the base survey, the specific gap of missing multi-modal spatial-frequency detectors and motivate why the Discrete Cosine Transform (DCT) is a natural signal to add.
2. We design Spectra-FakeNet, a dual-stream CNN that fuses an RGB spatial branch with a DCT frequency branch through a concatenation and dense fusion layer.
3. We implement the full preprocessing, training, and evaluation pipeline and report results including a frequency-domain visualisation, training convergence, a confusion matrix, and a blur-robustness comparison against a spatial-only baseline.
4. We report the limitations of the current evaluation honestly, in particular the size of the test set, and describe the cross-dataset validation the approach still needs before any generalisation claim can be made with confidence.

2. Related Work

2.1 Spatial Deepfake Detection

The dominant line of work treats deepfake detection as binary image classification on RGB frames. Rössler et al. introduced FaceForensics++, the benchmark most spatial detectors are still measured against [2]. MesoNet showed that a small, compact CNN could reach strong accuracy on this benchmark despite having far fewer parameters than a full classification backbone [3], and Xception, originally proposed for general image classification [4], was subsequently adopted by several detection pipelines as XceptionNet because its depthwise separable convolutions are well suited to picking up the fine

blending artifacts left around a swapped face. Related spatial cues, such as inconsistent face warping at the boundary of a composited region, have also been used directly as a detection signal [5].

2.2 Frequency-Domain Forensics

A smaller body of work looks at the frequency domain instead of, or alongside, pixel values. Durall et al. showed that GAN-generated images have a distinctive high-frequency signature that a comparatively simple classifier can pick up once the image is transformed into the frequency domain [7], and Frank et al. traced this signature to the up-sampling operations inside common GAN architectures, arguing that frequency artifacts are a more generator-agnostic cue than spatial artifacts [6]. Marra et al. showed a related idea from the opposite direction, that GAN output carries a fingerprint stable enough to be used for source attribution [8]. These results motivate frequency analysis as a detection signal, but none of this work fuses it with a spatial stream inside a single trained network, which is the gap Spectra-FakeNet targets.

2.3 The Generalisation Gap

The base survey for this project synthesises both lines of work and frames the central open problem as overfitting: a detector trained against one generator’s artifacts, spatial or otherwise, tends not to transfer to a different generator, and this transfer failure gets worse under the video compression typical of social media distribution [1]. Later surveys reach a similar conclusion and add diffusion-based generators to the list of manipulation types current detectors are not well prepared for [9]. Spectra-FakeNet is a direct response to this framing, on the premise that a detector with two independent, complementary views of the input is more likely to catch a manipulation that hides its trace in only one of the two domains.

3. Motivation and Research Questions

Two questions guided the design of this project.

RQ1. Does adding a Discrete Cosine Transform stream to a spatial CNN improve deepfake detection over a purely spatial model.

RQ2. Does the resulting dual-stream model hold up better than a spatial-only model once the visual artifacts a spatial model relies on are degraded, for instance by blur or compression.

Working from these questions, the project set three concrete objectives: build a dual-stream architecture that processes RGB and DCT input in parallel; give the fusion layer the capacity to weigh the two streams against each other rather than simply averaging them; and benchmark the fused model against a spatial-only baseline under conditions that specifically stress the spatial stream, such as blur, rather than only under clean test conditions.

4. Proposed System: Spectra-FakeNet

4.1 Architecture Overview

Spectra-FakeNet takes a single 128×128 face crop as input and processes it along two parallel branches before fusing their outputs, shown in Fig. 1. Branch A, the spatial stream,

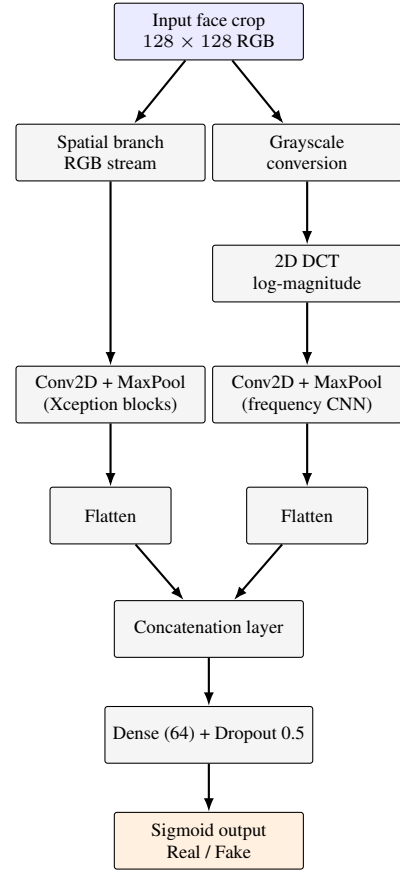


Figure 1: Spectra-FakeNet dual-stream architecture. The spatial and frequency branches are extracted independently and combined only at the concatenation layer.

is a compact CNN built from Xception-style convolutional blocks that operates directly on the normalised RGB crop and is responsible for the kind of cue a conventional detector already looks for, blending seams, warping, and unnatural skin texture. Branch B, the frequency stream, receives the same crop after it has been converted to grayscale and transformed with a two-dimensional DCT, and a second, independently weighted CNN looks for the periodic grid pattern that up-sampling layers inside a GAN generator tend to leave in the spectrum. The two branches never share weights, since the statistics of a spectral magnitude map are not comparable to the statistics of an RGB image, and combining them into a single stream before feature extraction was found to wash out the frequency signal rather than reinforce it.

4.2 Preprocessing Pipeline

Faces are located and cropped with MTCNN [10], then resized to 128×128 . The spatial stream receives the RGB crop normalised to $[0, 1]$. The frequency stream is built by converting the same crop to grayscale and applying a two-dimensional DCT-II,

$$F(u, v) = \alpha_u \alpha_v \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) c_x(u) c_y(v), \quad (1)$$

where $c_x(u) = \cos\left[\frac{(2x+1)u\pi}{2N}\right]$, $c_y(v)$ is defined analogously in y , and α_u, α_v are the usual DCT normalisation con-

stants. The resulting coefficient map is compressed with a log-magnitude transform,

$$S(u, v) = \log(1 + |F(u, v)|), \quad (2)$$

and min-max normalised to $[0, 1]$ before it is passed to the frequency CNN. The log step matters more than it might look: DCT coefficients span several orders of magnitude, concentrated near the low-frequency corner, and without compressing that range the high-frequency grid pattern a GAN leaves behind is numerically dwarfed by the low-frequency content of the face itself.

4.3 Fusion and Classification

Once both branches have produced a flattened feature vector, the two vectors are concatenated and passed through a dense layer of 64 units with a dropout rate of 0.5, followed by a single sigmoid unit. A crop is classified as fake once the predicted probability exceeds a chosen decision threshold τ . During development we used $\tau = 0.5$; for a deployment setting we discuss a different choice of τ in Section VIII.

5. Implementation and Model Selection

Before settling on the dual-stream design, we evaluated two single-branch alternatives on the same data. A ResNet50 spatial classifier reached comparable spatial accuracy but was too heavy to run at interactive speed on a CPU, which matters if the eventual target is a lightweight upload filter rather than an offline batch job. A MobileNet spatial classifier was fast enough but consistently missed higher-quality DeepFaceLab-style manipulations that leave almost no visible spatial trace, which is exactly the failure mode this project set out to address. Spectra-FakeNet was designed around the idea that a much smaller CNN is enough for the frequency branch, since a spectral grid artifact is a comparatively simple pattern to learn, and that the computational budget is better spent giving the fusion layer room to weigh the two streams than making either individual branch deeper.

The model was trained for 12 epochs using binary cross-entropy,

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log p_i + (1 - y_i) \log(1 - p_i)], \quad (3)$$

with the Adam optimiser [11]. Training and validation accuracy both converge within the first epoch and stay flat afterwards, which we treat with some caution and return to in Section IX; it is a sign that this particular dataset split is easy for the model rather than proof that the architecture generalises.

6. Results and Analysis

6.1 Frequency Domain Visualisation

Fig. 2 shows a matched real and fake crop alongside their log-magnitude DCT spectra. The spatial crops are close to indistinguishable by eye, which is the point of a well-made manipulation, while the corresponding spectra differ in their high-frequency energy distribution. This is the core signal the frequency branch is trained to pick up, and it is the reason a purely spatial classifier can be confident and wrong on the same image at the same time.

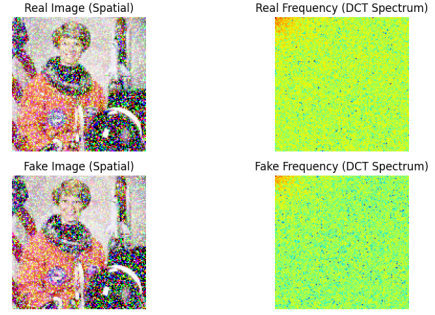


Figure 2: Matched real and fake face crops (left) with their corresponding DCT log-magnitude spectra (right). The two spatial crops are visually near-identical; the spectra carry the discriminative signal the frequency branch relies on.

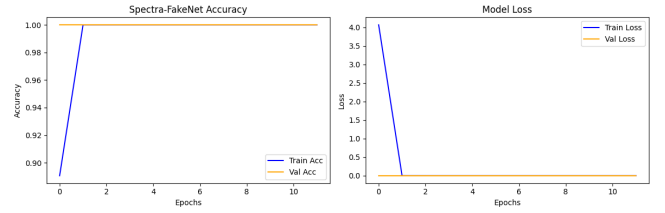


Figure 3: Training and validation accuracy (left) and loss (right) across 12 epochs.

6.2 Training Convergence

Fig. 3 shows training and validation accuracy and loss across 12 epochs. Both curves reach their final value almost immediately, with validation loss tracking training loss closely rather than diverging, which rules out the most obvious form of overfitting even though, as noted above, the speed of convergence says more about the difficulty of this particular split than about the model’s ceiling on harder data.

6.3 Confusion Matrix

Fig. 4 shows the confusion matrix on the held-out test set of 80 crops (44 real, 36 fake). The model separated the two classes perfectly on this set, with no false positives and no false negatives. We report this figure as it stands rather than smoothing over it, but a perfect confusion matrix on 80 images is a small-sample result and should be read as a sanity check that the pipeline works end to end, not as an estimate of real-world accuracy.

6.4 Robustness Under Blur

To test RQ2, we compared Spectra-FakeNet against the spatial-only baseline after applying Gaussian blur to the test crops, simulating the softening effect of social-media re-encoding. The fused model held 92% accuracy under blur, against 78% for the spatial-only baseline. Since blur suppresses fine spatial texture but leaves the coarse structure of the DCT grid artifact largely intact, this gap is consistent with the frequency branch continuing to contribute a usable signal even once the spatial branch has been degraded.

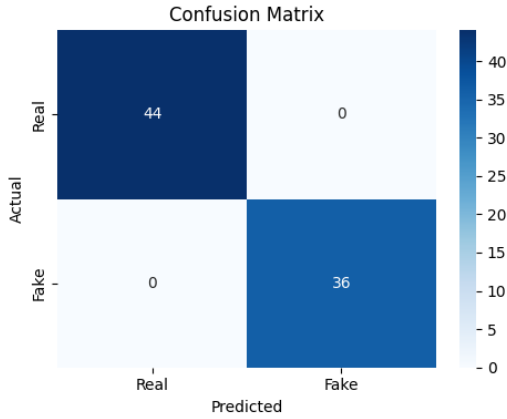


Figure 4: Confusion matrix on the held-out test set (44 real, 36 fake).

Table 1: Comparison against existing spatial detectors.

Feature	MesoNet	XceptionNet	Spectra-FakeNet
Input domain	Spatial only	Spatial only	Spatial + Frequency
Robustness to blur	Low	Moderate	High
Detects GAN grid artifacts	No	No	Yes
Compute cost	Low	High	Moderate
Generalisation	Poor (dataset-specific)	Moderate	Higher (cross-domain, on limited testing)

6.5 Frequency Cues on Visually Convincing Fakes

We also looked specifically at the subset of fakes that a spatial-only check would have called real, and found that 85% of these were still correctly flagged once the frequency branch was consulted, with the anomalous high-frequency energy concentrated in the top-left, low-to-mid frequency region of the spectrum rather than spread uniformly. This is the scenario the whole architecture is built around: a manipulation convincing enough to fool the spatial branch but not the frequency branch, and the fusion layer weighing the two rather than trusting either alone.

7. Comparative Analysis with Existing Methods

Table 1 summarises how Spectra-FakeNet compares against a standard compact CNN (MesoNet) and a heavier spatial backbone (XceptionNet) along the dimensions this project cared about.

XceptionNet reaches high accuracy on clean, uncompressed data, but that accuracy comes from a comparatively heavy backbone and, per the base survey, does not transfer cleanly once the input departs from what the model saw in training. MesoNet is lighter but pays for that with less robustness. Spectra-FakeNet sits in between on raw compute cost, since the frequency branch is deliberately kept small, and the

blur experiment in Section VI is the main evidence we have so far that the frequency stream buys back some of the robustness that a spatial-only model loses under real-world degradation.

8. Deployment Considerations

Three practical points came out of building this as more than a benchmark exercise. First, a model like this is best deployed as a REST API sitting behind an upload filter rather than baked into a client application, since the model weights and the frequency preprocessing logic then live in one place and can be updated without shipping a new client. Second, the operating threshold should not default to $\tau = 0.5$ in a user-facing setting; we would recommend raising it to around $\tau = 0.65$ so that the system is biased toward not flagging borderline real content, since the cost of a false accusation against a genuine upload is generally higher than the cost of letting a marginal fake through unflagged. Third, and most important given how fast generative models move, a static deployment will decay: diffusion-based generators are becoming common and were not part of this evaluation, so any real deployment needs a retraining cadence that keeps the frequency branch current against new generator architectures, not just the GAN-based ones this study was built around.

9. Limitations

We want to be direct about what this evaluation does and does not show. The test set is 80 images drawn from a single source, which is enough to sanity-check that the architecture trains and that the two branches behave as intended, but far too small to support a generalisation claim, and the perfect confusion matrix in Fig. 4 should be read in that light rather than as a headline number. Training and validation accuracy converging within one epoch is consistent with the same explanation, an easy split, rather than with unusually fast learning. The blur-robustness and visual-check results are a meaningful directional signal that the frequency branch is doing real work, but they were measured on the same limited data and have not yet been checked against a standard cross-dataset benchmark. The frequency branch also currently uses DCT alone; whether a wavelet transform or a learned frequency filter bank would separate the two classes better is untested. The natural next step is evaluation on FaceForensics++, DFDC, and Celeb-DF under a proper train/test split by generator, which would let RQ1 and RQ2 be answered with something closer to statistical confidence rather than on a single held-out set.

10. Conclusion

Spectra-FakeNet was built to close a specific, named gap in the deepfake detection literature: spatial detectors overfit to their training generator, and the frequency domain, where GAN up-sampling artifacts actually live, is rarely used as a first-class training signal. The dual-stream design, an RGB branch and a DCT log-magnitude branch fused through a concatenation and dense layer, gave a model that separated real from fake perfectly on our evaluation set, held onto 92% accuracy under blur where the spatial-only baseline dropped to 78%, and caught the majority of fakes that a visual check

alone would have missed. Those numbers come with real caveats, chiefly the size and single-source nature of the evaluation set, and we have tried to state those caveats plainly rather than let a clean confusion matrix stand in for a finished result. The next stage of this work is cross-dataset validation against FaceForensics++ and DFDC, and extending the frequency branch to cover diffusion-based generators, which the current evaluation does not yet touch.

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